

Case Study Report

Design and Development of a Patient-Specific Cranial Plate Using Medical Imaging and Additive Manufacturing Techniques

Case 03 — Large Parieto-Occipital Solid Plate

1. Introduction

1.1 Cranial Defect

Cranial defects are discontinuities in the skull that can occur due to trauma, decompressive craniectomy, tumor resection, congenital abnormalities, or infections. These defects compromise brain protection and can affect a patient's aesthetics and quality of life. Restoration of the cranial vault is achieved through a surgical procedure known as cranioplasty, using implant materials such as PMMA, titanium, or PEEK.

1.2 Need for Patient-Specific Implants (PSI)

Conventionally shaped implants often result in poor anatomical fit and longer surgical time. Patient-specific implants, designed directly from CT imaging data, accurately match the patient's own anatomy — leading to better fit and improved outcomes. This case study demonstrates a patient-specific implant design workflow using open-source software.

2. Objective

- To design a patient-specific cranial implant model from a segmented skull STL.
- To reconstruct a large parieto-occipital defect using the mirror-anatomy technique.
- To generate a continuous, solid (non-lattice) implant surface suitable for wide-area structural coverage.
- To demonstrate a complete, low-cost, open-source design workflow suitable for educational and research use.

3. Case Description

This case involved a large posterior cranial defect spanning the parieto-occipital region, visible as an open cavity in the skull model. Given the size of the defect, a solid (non-lattice) plate was designed instead of a perforated lattice structure, in order to provide continuous surface coverage across such a wide opening. The mirrored and smoothed

plate surface was fitted over the defect on the skull model, and the plate geometry was also isolated separately to check its shape and thickness before export.

4. Materials and Software Used

Category	Details
Imaging Data	Segmented skull STL model
Segmentation Software	3D Slicer (open-source)
Design Software	Autodesk Meshmixer (free)
Output Format	STL (Binary)
Intended Fabrication	FDM 3D Printing

5. Design Methodology — Mirror & Solid Surface Workflow

Step 1 — Import STL File → Import → load the segmented skull STL into Meshmixer.

Step 2 — Analysis → Inspector (Repair Skull) Open the Inspector tool from the Analysis panel to automatically detect and repair holes, non-manifold edges, and flipped normals, resulting in a clean, watertight mesh.

Step 3 — Duplicate Skull & Hide Original Use Edit → Duplicate to create a copy of the skull and hide the original, working only on the duplicate.

Step 4 — Enable Symmetry on Duplicate Skull Select the duplicate skull and enable the Symmetry option, setting the sagittal midline plane required for mirroring.

Step 5 — Select Defect Boundary & Separate Use the Sphere Brush to sketch along the large defect boundary spanning the parieto-occipital region. With Symmetry ON, the corresponding selection appears on the healthy opposite side.

Step 6 — Mirror the Separated Patch Isolate the healthy mirrored region using Edit → Separate, then apply Analysis → Mirror (or Edit → Mirror) across the sagittal axis to generate a patch shaped for the large defect area.

Step 7 — Show Original Skull & Align Patch Make the original skull visible again and use the Transform tool to align the mirrored patch onto the actual defect site.

Step 8 — Smooth the Mirrored Surface Given the size of the defect, apply Laplacian/Sculpt smoothing across the mirrored patch to remove any stepping artifacts and ensure a continuous, even surface.

Step 9 — Refine Margins Against the Skull Use the Sculpt → Refine brush along the plate edges to blend the margins cleanly into the surrounding bone, ensuring a flush fit given the wide contact area.

Step 10 — Keep Solid Surface (No Lattice Conversion) Unlike smaller defects, this plate was intentionally kept as a continuous solid surface rather than converted into a lattice pattern, to maintain full structural coverage over the large opening.

Step 11 — Isolate and Inspect Plate Geometry Separate the finished plate from the skull model and inspect it independently to verify its shape, curvature, and consistent thickness across the entire surface.

Step 12 — Verify Fit on the Skull Place the isolated solid plate back onto the skull model and confirm it covers the defect fully with no visible gaps at the margins.

Step 13 — Export STL Run Analysis → Inspector one final time to confirm mesh integrity, then export the finished solid plate as a binary STL file for printing/further use.

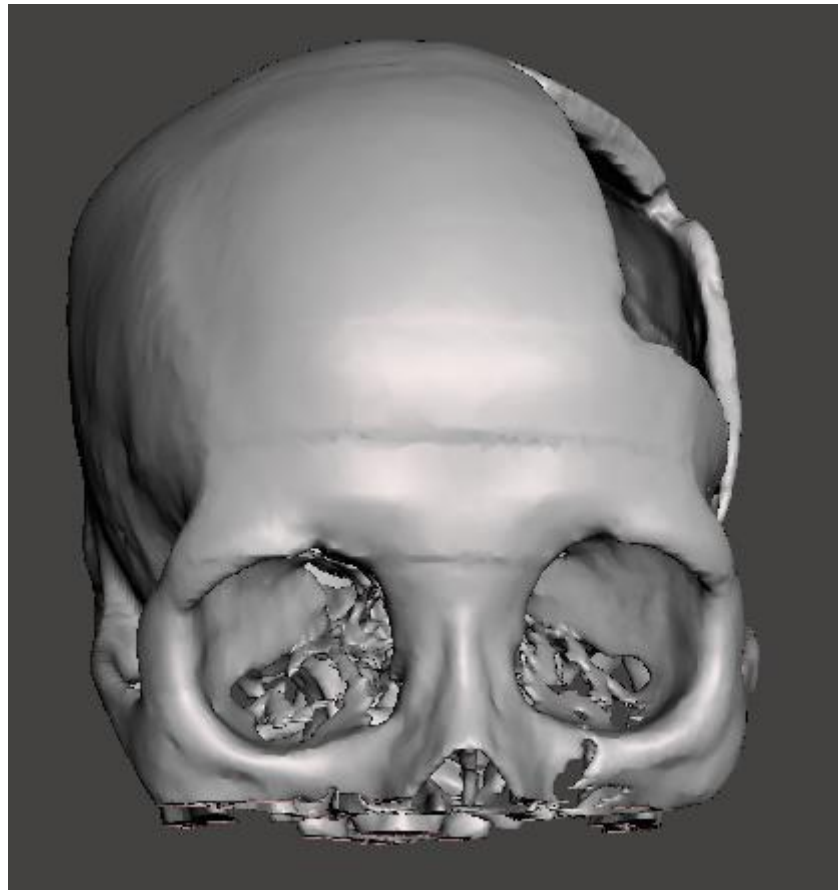
6. Results

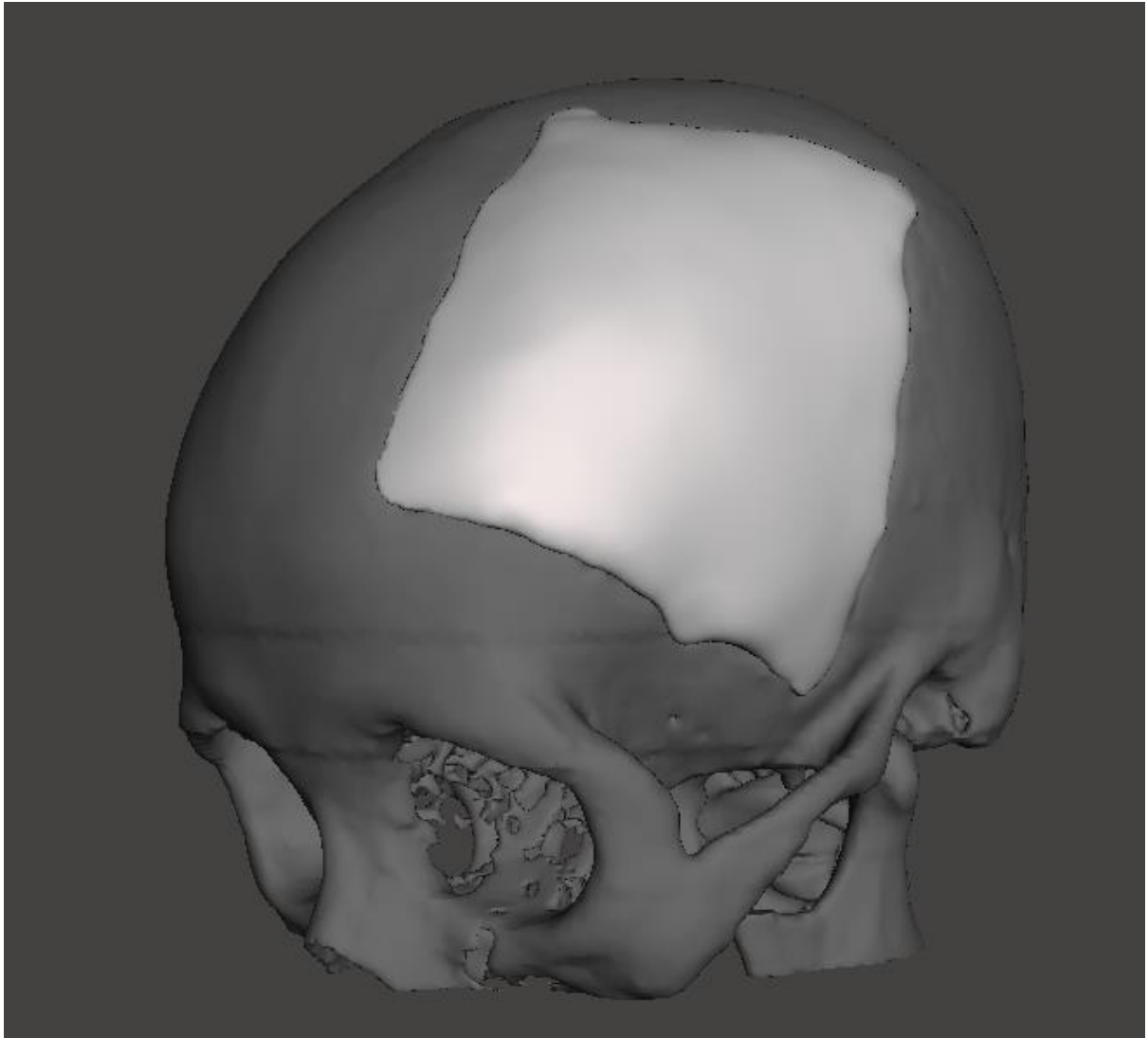
The completed design is a patient-specific, continuous solid cranial plate covering the large parieto-occipital defect region. The plate was smoothed at the margins for a flush fit and verified both while positioned on the skull model and as an isolated, independent geometry.

Parameter	Value
Defect Region	Parieto-Occipital
Design Technique	Mirror anatomy + Solid surface (non-lattice)
Coverage	Wide / continuous
Output File	STL (Binary)
Software Used	Meshmixer, 3D Slicer

7. Conclusion

This case demonstrates that for larger cranial defects, a continuous solid plate can be a more suitable design choice than a lattice structure, prioritizing full structural coverage over weight reduction. The mirror-anatomy technique remained effective even at this larger scale, producing an anatomically-matched plate that was verified both in position and in isolation before export.





Disclaimer: This case study is created purely for **educational and research purposes** as part of the Cranial3DLab project. It is intended to demonstrate the technical design workflow using open-source tools and does not represent a validated clinical implant. Any real-world surgical implant requires professional medical oversight, regulatory approval, and certified manufacturing processes.